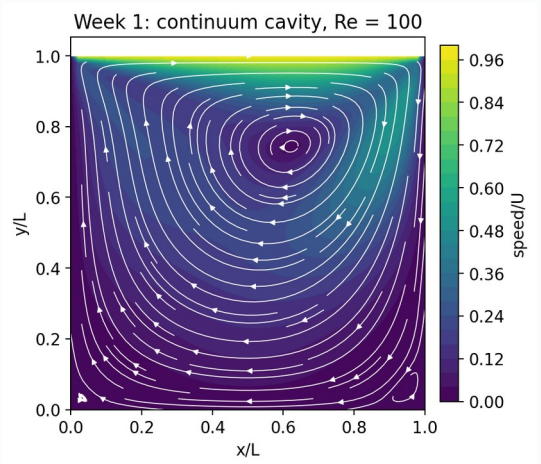
AI in Fluid Dynamics

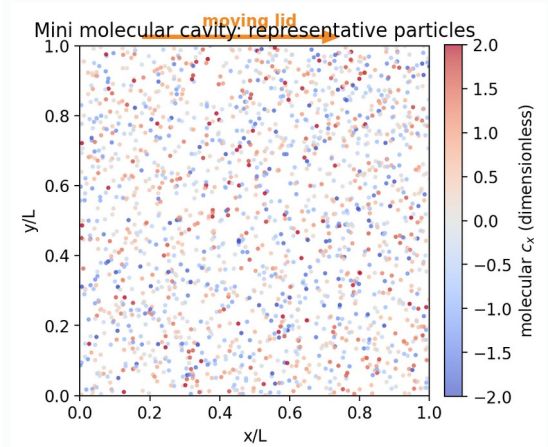
Week 3

From Continuum Cavity to Molecular Cavity

Kinetic theory, Boltzmann equation, DSMC sampling, and data quality for machine learning



Week 1: continuum field



Week 3: molecular samples

same geometry
different description

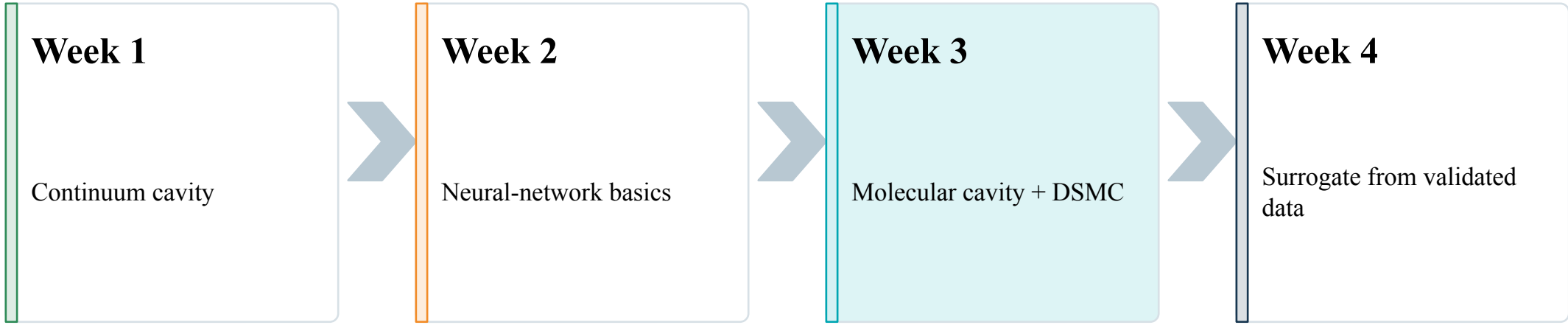
Physics -> solver -> validated data -> AI surrogate

Instructor: Ehsan Roohi | M&I-ENG 690A

The course story continues

COURSE STORY

Week 3 is not a separate kinetic-theory detour; it creates the data that later AI models will learn.



Central question

How do we turn physically meaningful molecular simulations into trustworthy training data - without letting statistical noise or numerical bias become “ground truth”?

The geometry is continuous across the course; the modeling description and validation questions become richer.

Learning outcomes

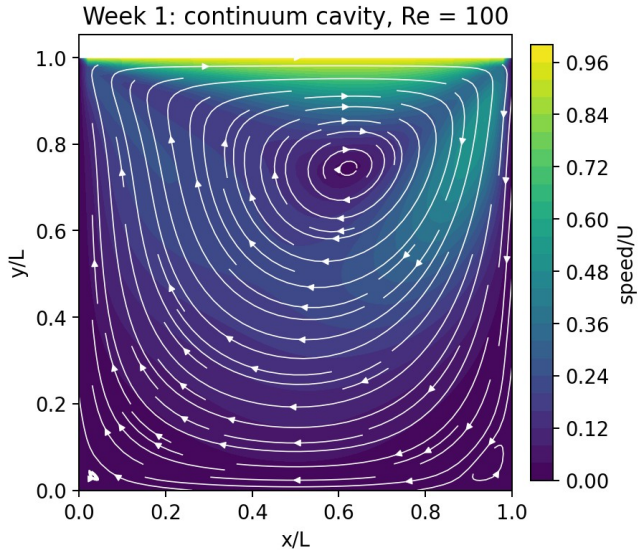
OVERVIEW

1. Interpret a molecular velocity distribution and a Maxwellian equilibrium.
2. Recover density, bulk velocity, temperature, pressure, and heat flux from particles.
3. Explain the Boltzmann equation and why DSMC is used.
4. Trace the DSMC cycle: move, walls, cells, collisions, sampling.
5. Quantify statistical noise and connect data quality to ML validation.

By the end of the week, students should see DSMC output as a statistical dataset with physics, metadata, and uncertainty.

Same cavity, different modeling language

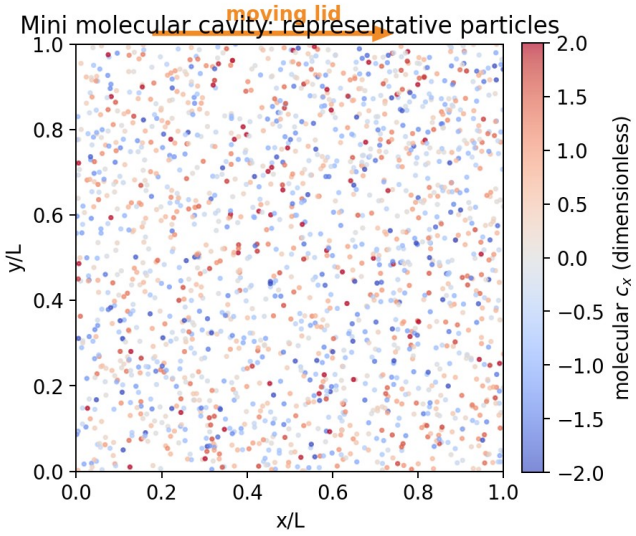
PHYSICS BRIDGE



Continuum

What changes?

State	u, v, p on a grid	particle positions + molecular velocities
Wall model	no-slip / prescribed T	stochastic gas-surface reflection
Control	Reynolds number	Knudsen number
Output	smooth deterministic field	noisy sampled field
Validation	residual + Ghia benchmark	statistics + DSMC criteria + paper data



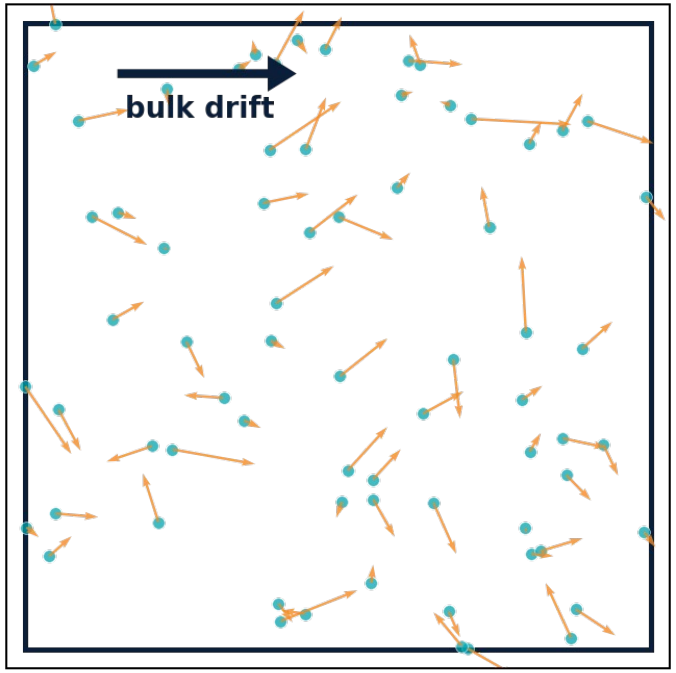
Molecular / kinetic

The Week-3 question is not “which picture looks nicer?” It is “which model is valid in this regime, and how trustworthy are its samples?”

A macroscopic velocity hides many molecular velocities

KINETIC VIEW

One macroscopic velocity hides many molecular velocities



Molecular velocity, c

The instantaneous velocity of one molecule. It contains organized motion plus thermal fluctuation.

Bulk velocity, U

The average molecular velocity in a small region and time window.

Thermal fluctuation, c - U

The random part used to define temperature, stress, and heat flux.

A low-speed gas flow can have a small U even though individual molecules move hundreds of meters per second.

The distribution function is the kinetic state variable

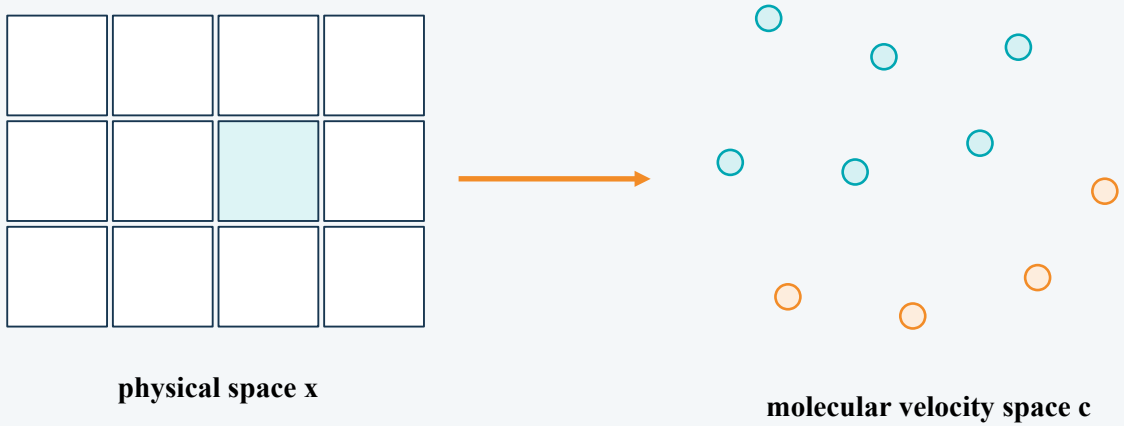
KINETIC THEORY

$$f(\mathbf{x}, \mathbf{c}, t)$$

What does f describe?

How molecular populations are distributed over physical position x, molecular velocity c, and time t.

Phase space: physical cells + velocity space



One continuum value U(x) is replaced by an entire local distribution over c.

Kinetic theory keeps information that is compressed away when we store only density, velocity, and temperature.

Fluid variables are moments of the distribution

KINETIC THEORY

Density

n = integral f dc
How many molecules?

Velocity

U = <c>
What is the average motion?

Temperature

T proportional to <|c-U|^2>
How strong are thermal fluctuations?

Pressure

p = n k_B T at equilibrium
How large is isotropic normal stress?

Heat flux

q proportional to <|c-U|^2(c-U)>
How is thermal energy transported?

DSMC does not output a smooth field directly. It outputs particle histories; macroscopic fields are estimators built from those histories.

Maxwellian equilibrium: the reference distribution

KINETIC THEORY

$$f_M = n\left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{m|\mathbf{c}-\mathbf{U}|^2}{2k_B T}\right]$$

Center

Set by the bulk velocity U.

Width

Set by temperature and molecular mass.

Amplitude

Set by number density n.

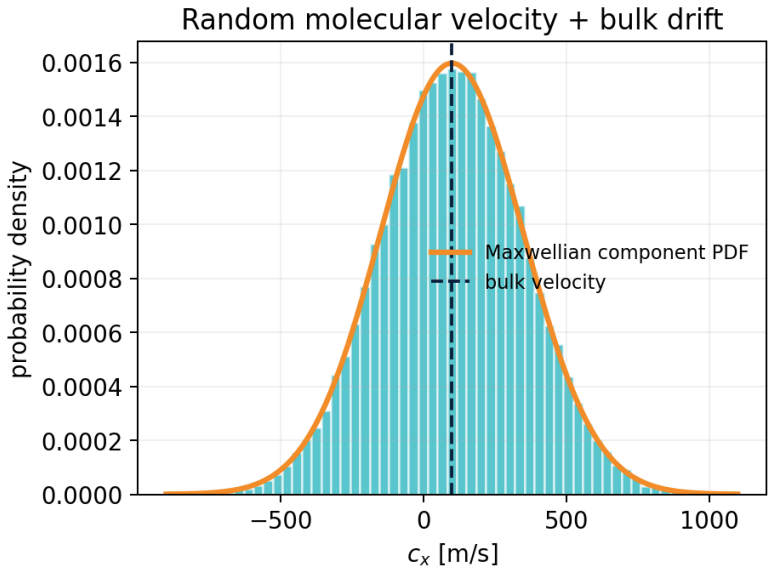
Equilibrium does not mean “molecules stop moving”

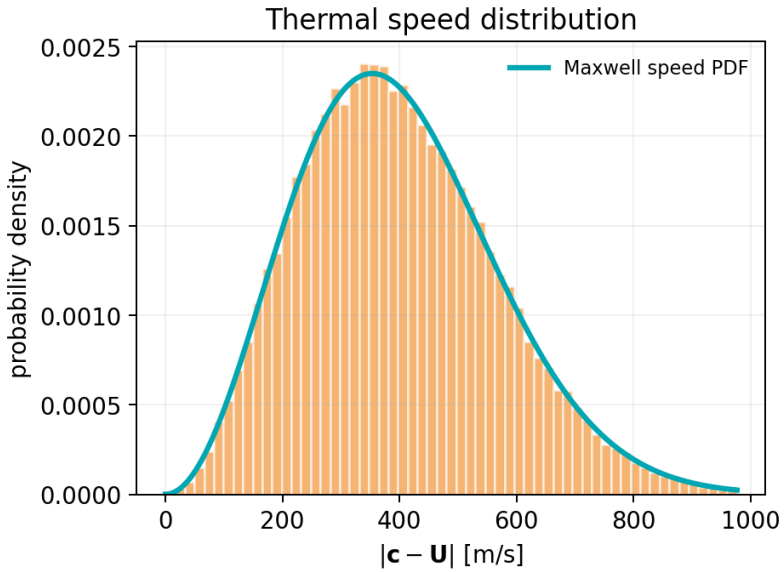
It means the velocity statistics have a specific form and there is no unresolved directional imbalance in the local rest frame.

Near walls, shocks, or strong gradients, the local distribution can depart from a Maxwellian even when density, U, and T look smooth.

Sampling a Maxwellian: what the notebook will do

LAB 1





Given

Argon, T = 300 K, and a prescribed bulk drift.

Generate

Three Gaussian velocity components for each particle.

Recover

Sample U, T, p, and q from finite particle moments.

The histogram is not the result to memorize; the connection between distribution parameters and macroscopic moments is the learning goal.

Finite-particle estimates are random variables

SAMPLING

Bulk velocity estimator

Average all molecular velocities in the sample.

Temperature estimator

Average fluctuation energy after subtracting the sampled bulk velocity.

Heat-flux estimator

Average a higher-order, signed thermal-energy flux. It is especially noisy.

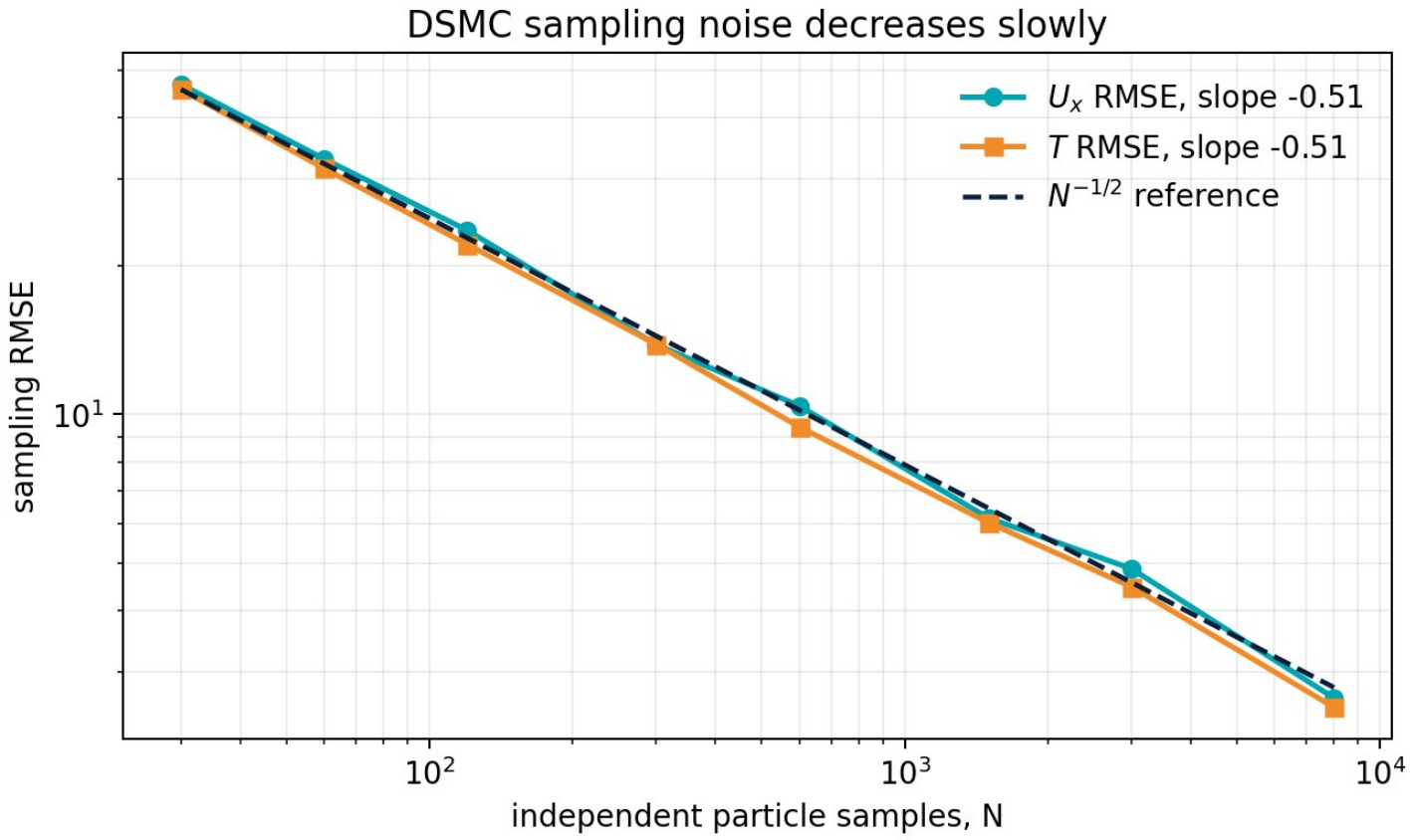
One possible finite sample

quantity	imposed	estimated	interpretation
Ux	100 m/s	near 100	mean drift
Uy, Uz	0	small nonzero	finite-sample fluctuation
T	300 K	near 300	thermal variance
q	0 at equilibrium	small nonzero	higher-order noise

A correct estimator can still be noisy; repeated samples produce a distribution of estimates.

Statistical error falls approximately as 1/sqrt(N)

SAMPLING



Core scaling

10x smaller standard error needs about 100x more independent samples.

Low-speed difficulty

The thermal molecular speed may be much larger than the small bulk-flow signal.

ML consequence

Noise creates a label-error floor that a network cannot eliminate by architecture alone.

The raw number of stored samples is not always the effective number of independent samples.

Effective sample size is more than particle count

SAMPLING

Particles per cell

More simultaneous observations

Number of cells

Spatial resolution and local count

Sampling duration

More time observations after transients

Sampling correlation

Reduces independent information

Repeated random seeds

Estimates run-to-run uncertainty

Long averaging helps only after the transient has decayed and only when the samples add new information.

Noise is not the only data-quality problem

SAMPLING

Variance / statistical noise

Reduced by more particles, longer averaging, multiple seeds, and appropriate estimators.

Visible symptom: rough fields and seed-to-seed scatter.

Bias / numerical or model error

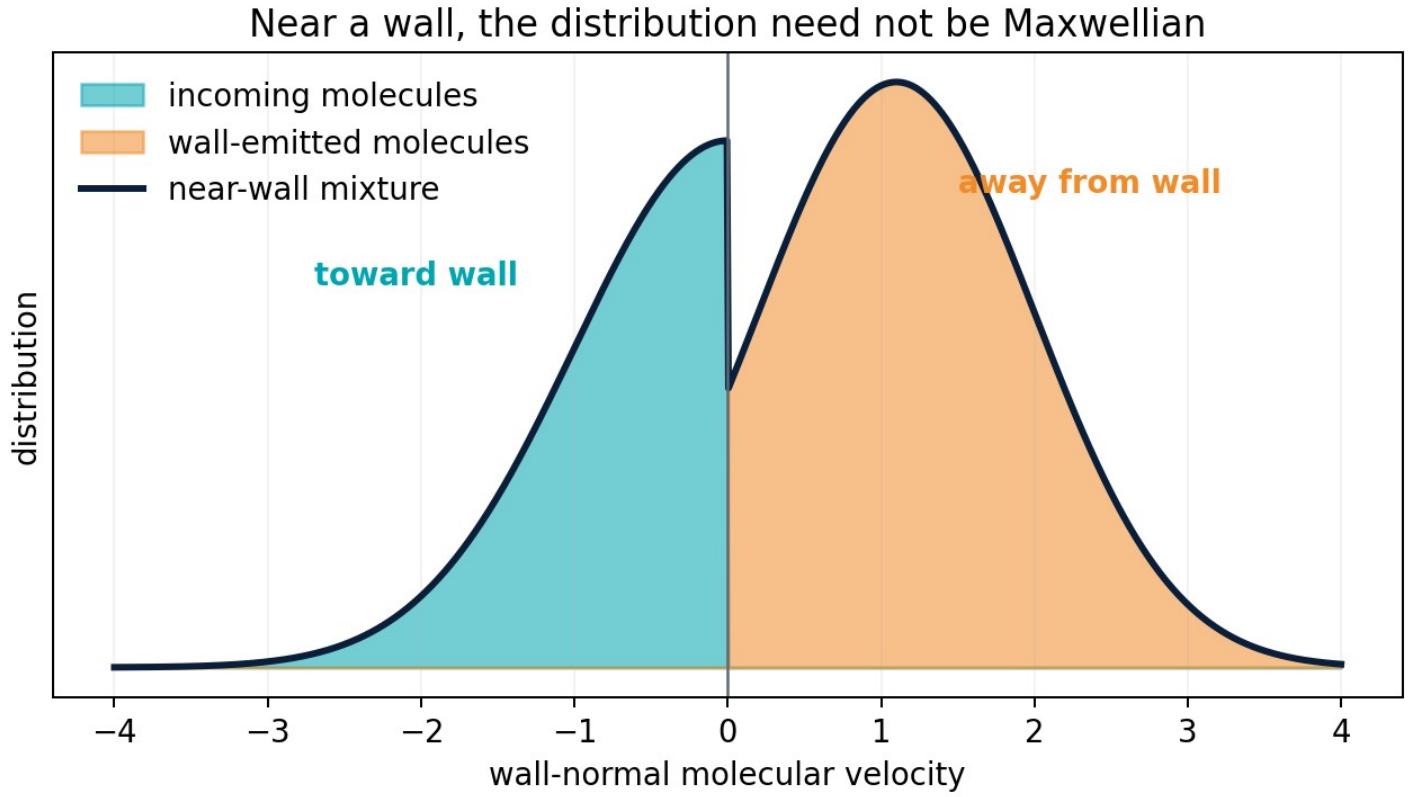
Caused by cells that are too large, time steps that are too large, incorrect boundaries, an unsuitable collision model, or unresolved gradients.

Averaging does not remove it.

A smooth, converged average can still be systematically wrong. Independence and validation remain essential.

Near a wall, the distribution need not be Maxwellian

NON-EQUILIBRIUM



Incoming population

Carries information from the gas toward the wall.

Outgoing population

Carries the wall temperature, velocity, and reflection model.

Macroscopic signatures

Velocity slip, temperature jump, Knudsen layers, and anisotropic stress/heat flux.

Wall conditions are not a minor implementation detail; they directly shape the kinetic distribution and therefore the training data.

The Boltzmann equation: transport plus collisions

BOLTZMANN

$$\frac{\partial f}{\partial t} + \mathbf{c} \cdot \nabla_{\mathbf{x}} f + \mathbf{a} \cdot \nabla_{\mathbf{c}} f = Q(f, f)$$

Left side: transport

Molecules move through physical space and, if forces exist, through velocity space.

Right side: collisions

The nonlinear collision operator redistributes molecular velocities while preserving collision invariants.

Why we care

The equation contains the complete kinetic state, and its moments recover continuum quantities.

But direct solution is expensive - especially for many parameter cases needed by AI.

DSMC is a stochastic particle method designed to reproduce the Boltzmann dynamics of dilute gases.

Why the Boltzmann equation is difficult to solve directly

BOLTZMANN

High dimension

3 physical + 3 velocity coordinates + time

Nonlinear collision integral

Pre- and post-collision states are coupled

Boundary physics

Diffuse/specular reflection and accommodation

Multi-scale structure

Bulk flow, Knudsen layers, shocks, corners

Large parametric cost

Many Kn, wall, geometry, and gas cases for design

AI opportunity

Learn mappings only after solver data are validated

The high cost of repeated kinetic solutions motivates surrogates; the high complexity also makes extrapolation dangerous.

DSMC separates molecular movement and collisions

DSMC

Transport substep

Particles move ballistically during a short time interval. Wall crossings are processed with a gas-surface model.

Collision substep

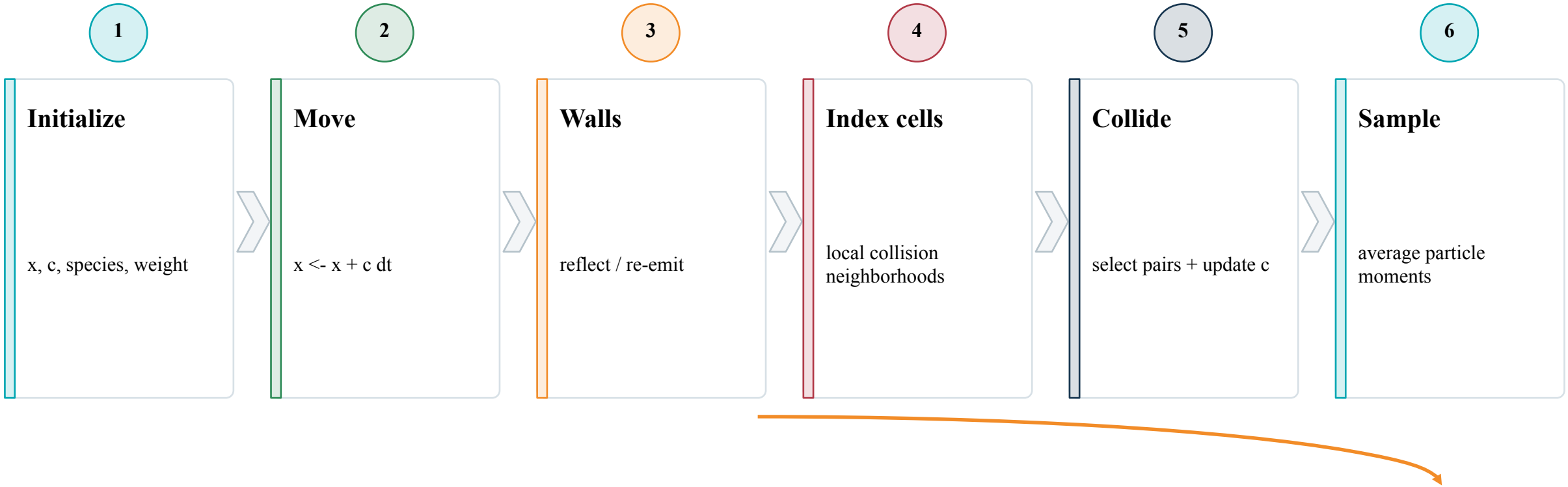
Pairs are selected locally inside cells and their velocities are updated stochastically.

repeat for many time steps

The time step must be short enough that the split transport/collision picture remains accurate.

The DSMC cycle

DSMC ALGORITHM



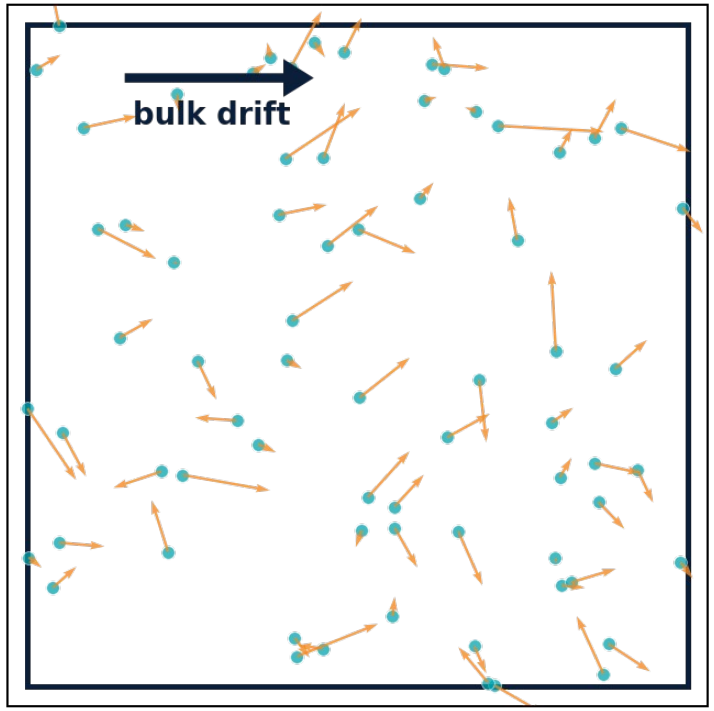
Continue until transients decay and the desired statistical precision is reached

The Week-3 notebook exposes every stage; the later Project-1 code adds VHS, NTC, GPU acceleration, validation, and independence studies.

Step 1 - initialize representative particles

DSMC ALGORITHM

One macroscopic velocity hides many molecular velocities



Positions

Usually sampled throughout the physical domain according to the initial density.

Velocities

Sampled from a Maxwellian or another prescribed distribution.

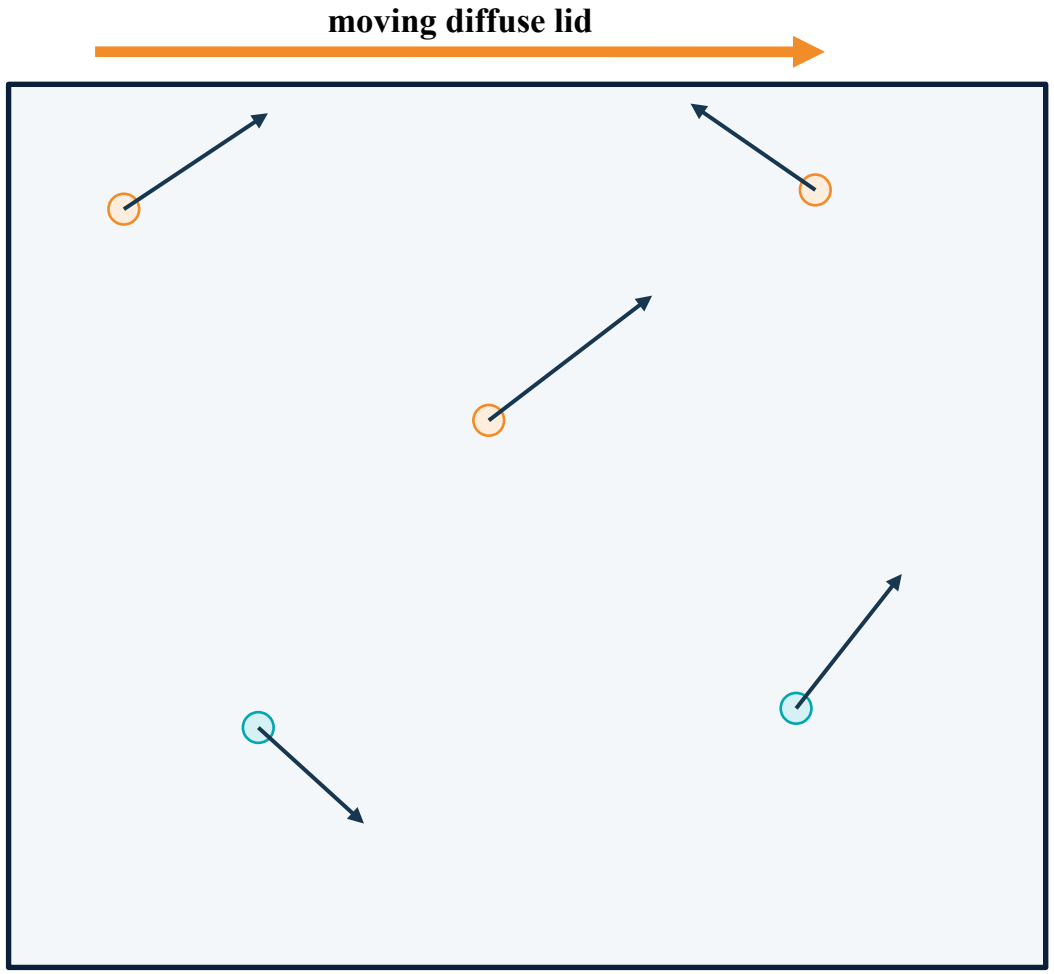
Statistical weight

One simulated particle represents many real molecules.

DSMC tracks representative particles, not every physical molecule.

Step 2 - move particles and apply wall physics

DSMC ALGORITHM



Ballistic move

Update positions with the current molecular velocity during dt.

Diffuse reflection

Re-emitted velocity is sampled from the wall temperature and wall velocity.

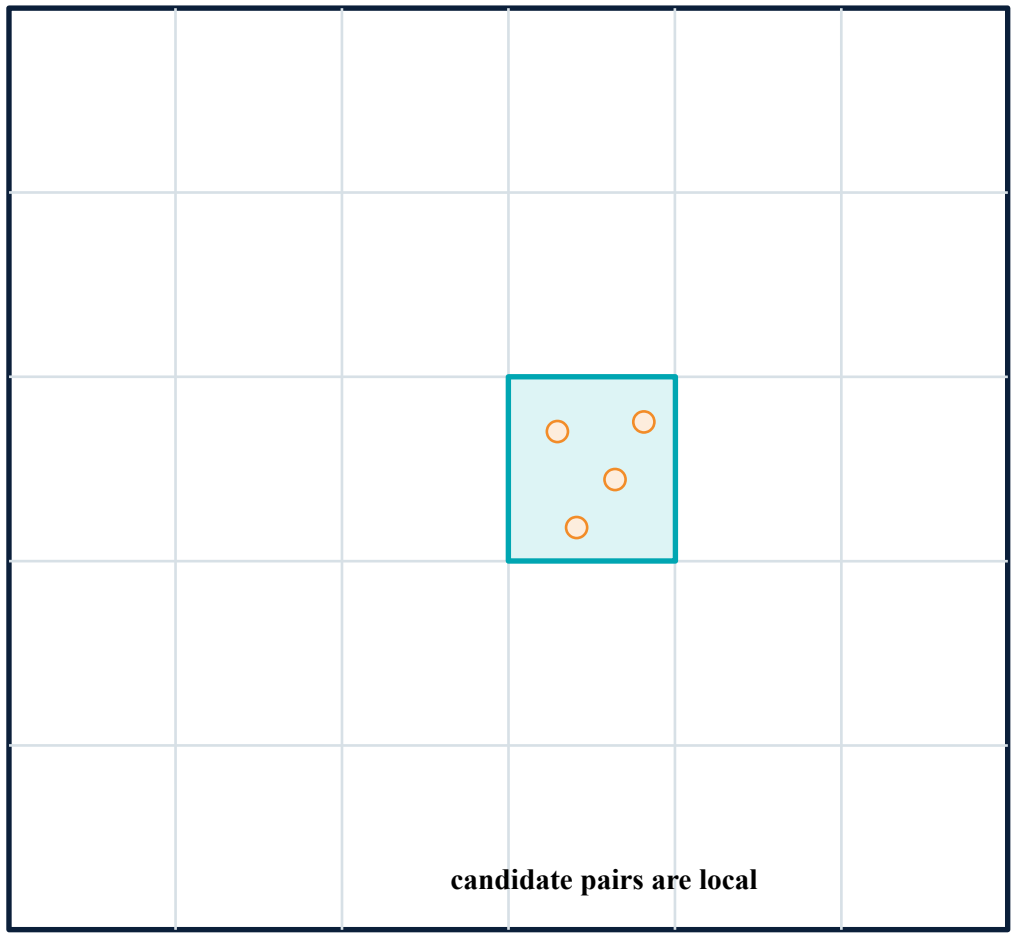
Kinetic consequence

The top lid injects tangential momentum; stationary walls remove or redistribute it.

Gas-surface interaction parameters are physics inputs and should later be retained as ML features or metadata.

Step 3 - collision cells enforce locality

DSMC ALGORITHM



Cell-size criterion

Cells should generally resolve the mean free path and important gradients.

$$\Delta x \lesssim \lambda$$

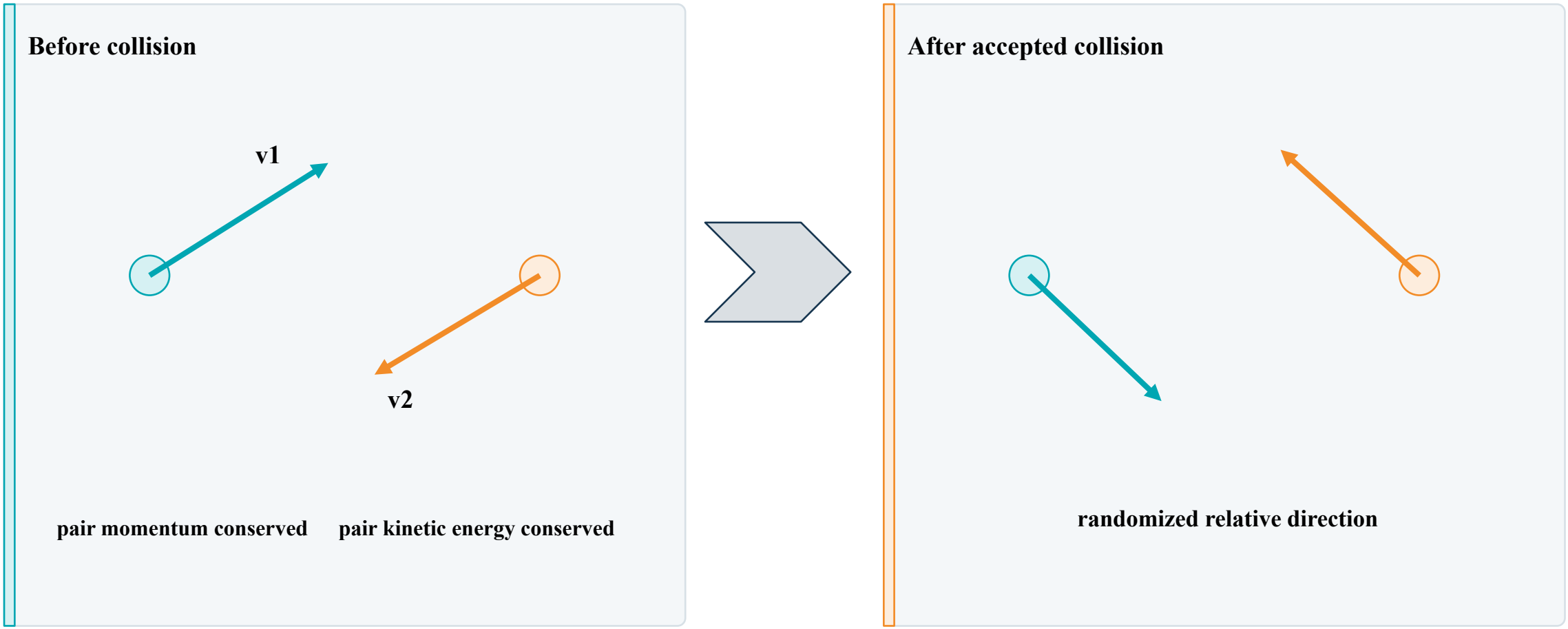
Why it matters

If a cell is too large, particles that are far apart may be treated as collision partners, creating nonlocal bias.

At low Kn, resolving the small mean free path makes DSMC expensive - one reason AI acceleration is attractive.

Step 4 - stochastic collisions preserve invariants

DSMC ALGORITHM

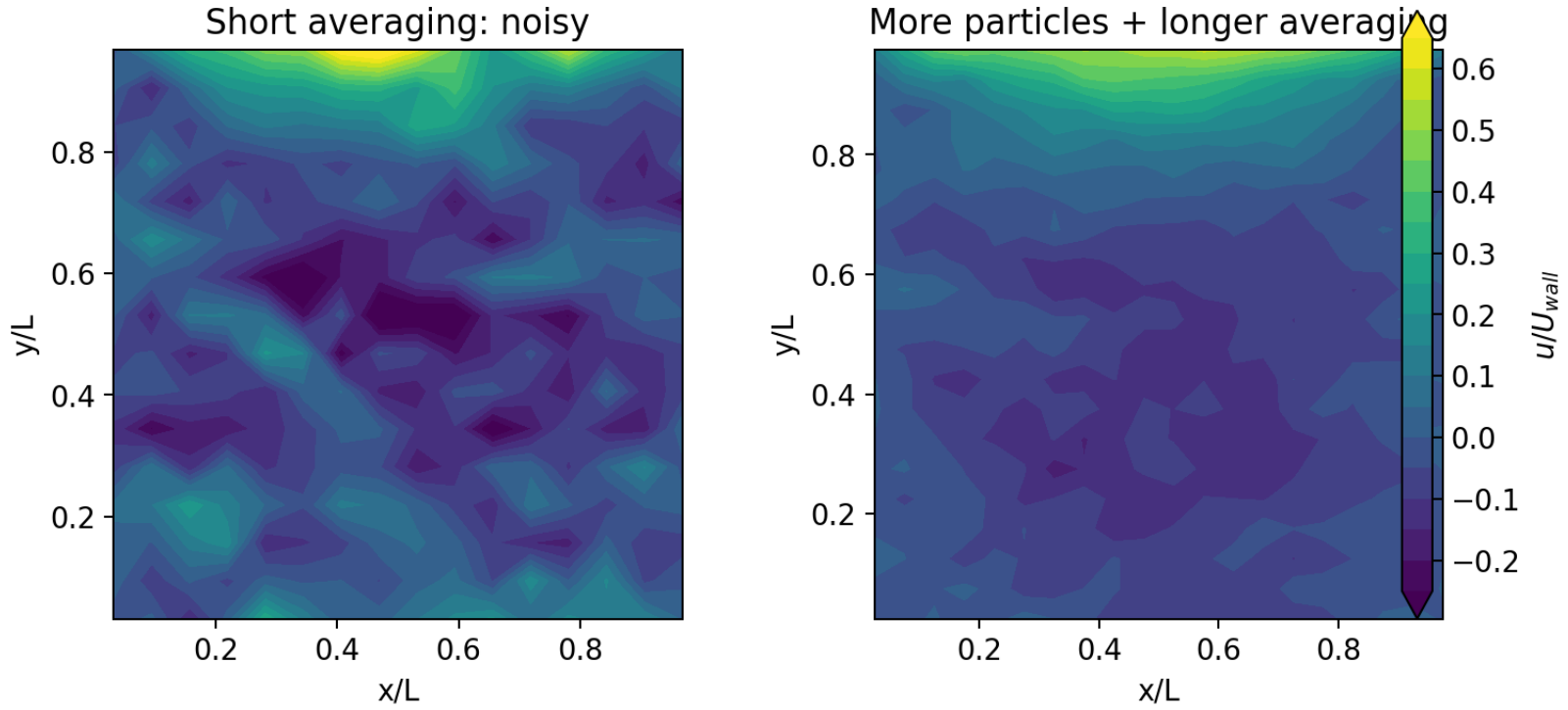


The Week-3 teaching model uses a simplified local elastic-pair rule; Project 1 uses VHS collision rates and No-Time-Counter selection.

Step 5 - sample, average, and attach uncertainty

DSMC ALGORITHM

The flow signal is the same idea; the statistical quality is not



Instantaneous field

Very noisy because each cell has a finite particle population.

Time-averaged field

Lower variance after transients, if samples add independent information.

Dataset record

Store mean values plus PPC, number of samples, seed, grid, dt, and model settings.

Never hide unsmoothed data or omit the sampling settings used to produce a “clean” contour.

DSMC accuracy criteria become AI data-quality criteria

DATA QUALITY

Cell size

Resolve mean free path and gradients
Diagnostic: dx/lambda

Time step

Resolve collision and transport time scales
Diagnostic: dt/tau

Particles/cell

Control local estimator variance
Diagnostic: PPC

Transient

Do not average the startup response
Diagnostic: sample_start

Sampling

Use a sufficient averaging window
Diagnostic: N samples

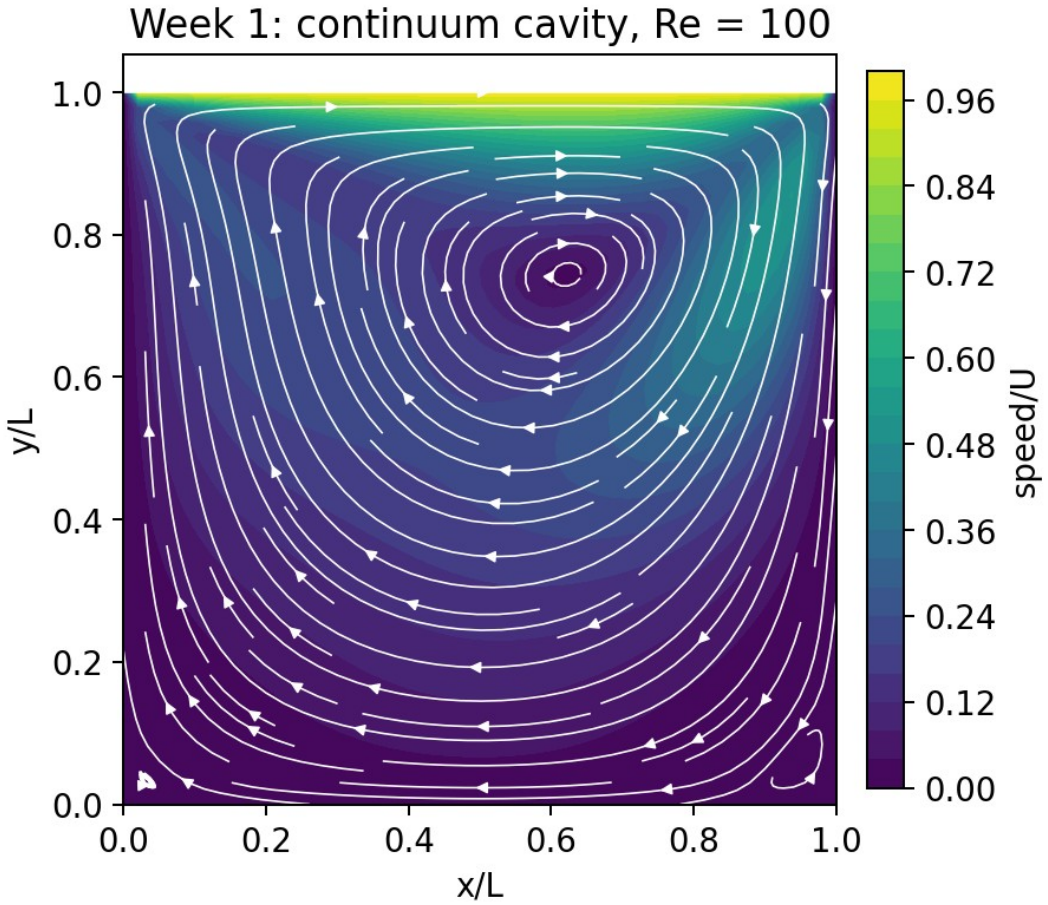
Seeds

Measure stochastic repeatability
Diagnostic: seed spread

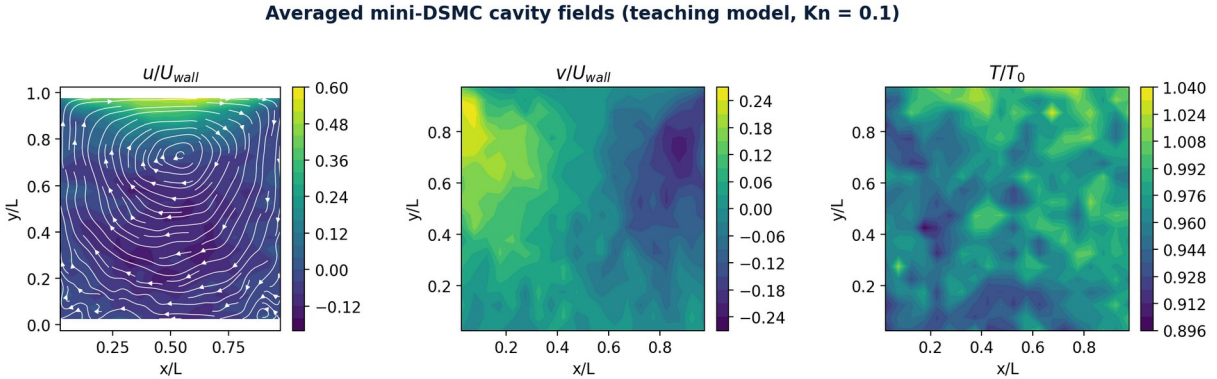
A polished neural-network prediction cannot be more trustworthy than the simulation data and metadata used to train it.

Revisit the cavity: Week 1 and Week 3 side by side

CAVITY STORY



Continuum CFD: smooth deterministic field



Mini molecular model: averaged but noisy field

Same lid-driven mechanism, but molecular wall slip, stochastic sampling, and kinetic validity now enter the interpretation.

What should a rarefied molecular cavity reveal?

CAVITY PHYSICS

Velocity slip

Gas adjacent to the moving lid does not exactly match the wall speed.

Temperature jump

Gas temperature next to a wall can differ from wall temperature.

Primary recirculation

Lid momentum is transferred through collisions and wall interactions.

Corner non-equilibrium

Incoming and wall-emitted populations can be strongly different.

Kn-dependent structure

Vortex location, slip, and thermal fields change with rarefaction.

Statistical noise

Small signals require particles, time averaging, and repeatability checks.

A plausible vortex picture is not sufficient validation; students must separate physical trend, sampling noise, and model limitations.

Week 3 uses a mini cavity - deliberately

LAB 2

Required this week

- Run a fast CPU teaching model
- Trace move, walls, cells, collisions, sampling
- Compare short and better averaging
- Export one ML-ready table
- Explain limitations honestly

Reserved for Project 1

- GPU VHS + NTC solver
- Paper validation
- Grid, PPC, dt, and sampling independence
- Multiple Kn cases
- Vortex-Kn trends and final dataset

This scope keeps Week 3 conceptually strong without turning one weekly exercise into the full research project.

A DSMC field becomes a supervised-learning table

AI BRIDGE

One row = one grid location in one validated case

log10 Kn	Uwall/c_th	x/L	y/L	rho/rho0	u/Uwall	v/Uwall	T/T0
-1.0	0.38	0.025	0.025	1.02	-0.04	0.01	0.98
-1.0	0.38	0.075	0.025	1.01	-0.03	0.02	0.99
...	...	...	...	...	...	...	...

features / coordinates **targets**

Pointwise surrogate

Inputs: log Kn, wall conditions, x/L, y/L. Outputs: rho, u, v, T.

Case-level surrogate

Inputs: Kn and wall parameters. Outputs: vortex center or integrated quantities.

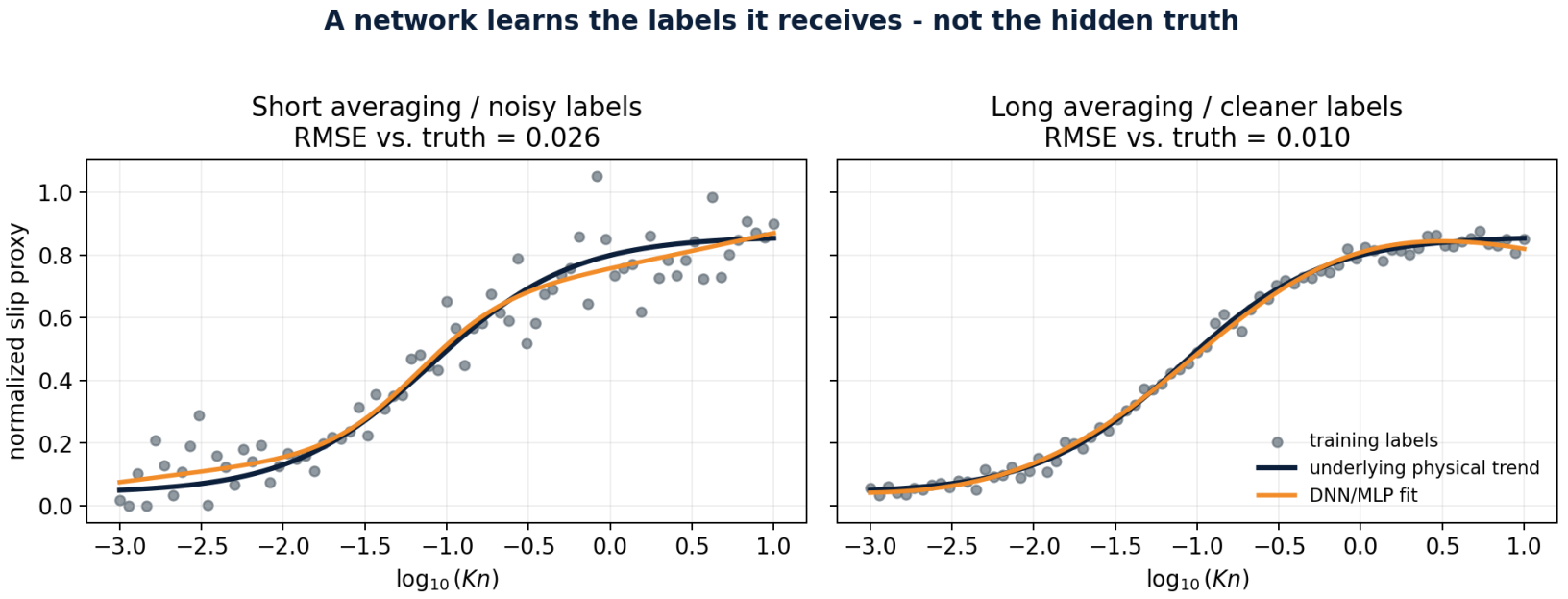
Metadata

Grid, PPC, dt, sample count, seed, collision model, wall model, uncertainty.

Do not let the ML table erase the numerical provenance of each target value.

A network learns the labels it receives - not hidden truth

AI + NOISE



Training loss

Measures agreement with the supplied labels.

Source-data error

Can be statistical noise or numerical bias that the loss never sees.

Correct response

Improve/quantify the data and validate by cases, regimes, and physics.

A smooth neural curve is not evidence that the DSMC labels were converged.

How to validate an ML surrogate trained on DSMC data

AI VALIDATION

Case-wise split

Hold out entire Kn or wall-condition cases, not random points from the same field.

Regime-wise error

Report performance separately in slip, transition, and high-Kn regimes.

Wall-region checks

Evaluate slip, jump, and near-wall errors, not only global RMSE.

Seed repeatability

Compare predictions with DSMC seed-to-seed uncertainty.

Physical checks

Positivity, symmetry, conservation trends, sensible wall behavior.

Extrapolation test

Deliberately evaluate outside the training Kn range and state failure.

Randomly splitting grid points from one field can produce severe leakage and falsely optimistic test error.

Week 3 is the on-ramp to Project 1

ROADMAP

Week 3 Lab 1

Maxwellian moments + noise scaling

Week 3 Lab 2

Mini cavity + ML-ready table

Project 1A

GPU preflight + baseline DSMC

Project 1B

Paper validation + independence

Project 1C

Multi-Kn dataset + vortex trend

Later AI

DNN, GPR, DeepONet / operator surrogate

The full project philosophy remains: physics -> DSMC solver -> validation -> independence -> dataset -> AI surrogate.

Week 3 submission: exactly three files

ASSIGNMENT

1

Lastname_Firstname_Week3_Lab1.ipynb

Completed Maxwellian, noise, and ML-label notebook with visible outputs.

2

Lastname_Firstname_Week3_Lab2.ipynb

Completed mini-cavity notebook with visible outputs and exported dataset.

3

Lastname_Firstname_Week3_Report.pdf

One report containing every written answer, requested table, figure, and interpretation.

Incomplete submission

Notebook only = incomplete. PDF only = incomplete. Written answers belong in the PDF; code and outputs remain in the notebooks.

What students will do this week

ASSIGNMENT

Lab 1 - Maxwellian, noise, ML

1. Sample argon velocities
2. Estimate U, T, p, and q
3. Repeat at several particle counts
4. Fit the error slope
5. Train the same small MLP on noisy and cleaner labels
6. Explain why low loss is not data validation

Lab 2 - Mini molecular cavity

1. Generate the DSMC flowchart
2. Verify collision conservation
3. Run a quick case
4. Run a better-sampled case
5. Compare fields and centerline profiles
6. Export an ML-ready table
7. State what the mini model cannot prove

The notebooks are scaffolded and CPU-friendly; the intellectual work is interpretation, data-quality reasoning, and clear reporting.

Week 3 summary

SUMMARY

1	**Kinetic state**	A distribution of molecular velocities, not only a few macroscopic fields.
2	**DSMC**	A stochastic approximation to dilute-gas kinetic dynamics using move, wall, cell, collision, and sampling steps.
3	**Data quality**	Statistical noise decreases slowly; numerical bias needs separate independence checks.
4	**Cavity story**	The same geometry now exposes slip, jump, wall non-equilibrium, and noisy sampling.
5	**AI message**	Validate the solver and uncertainty before asking a network to learn the data.

Exit question

What information can be lost when a noisy molecular distribution is reduced to a single averaged field - and then compressed again into an ML model?

Worked finite-N temperature estimate

Assume N independent, equally weighted 3D molecular velocities at one temperature. Estimate the bulk velocity from the same sample.

For N = 2, use c1 = (−1, 0, 0) m/s and c2 = (1, 0, 0) m/s. Let $m/k_B = 1$ K s²/m².

The sample mean is Ubar = (0, 0, 0) m/s. The summed fluctuation energy is $\Sigma|c_i - \bar{U}|^2 = 1 + 1 = 2$ m²/s².

The usual moment estimate is $T_N = (m/k_B)\,\Sigma|c_i - \bar{U}|^2/(3N) = 1/3$ K.

The Bessel-corrected estimate is $T_{est} = (m/k_B)\,\Sigma|c_i - \bar{U}|^2/[3(N-1)] = 2/3$ K.

This arithmetic sample illustrates the estimators. It does not establish the true temperature from two particles.

Why the finite-N correction matters

For fixed N iid samples with finite variance, subtracting the estimated mean uses one degree of freedom per velocity component.

$E[\Sigma|c_i - \bar{U}|^2] = 3(N-1)k_B T/m$, so $E[T_N] = (N-1)T/N$.

At N = 8 the expectation is 0.875T; at N = 24 it is about 0.95833T. The corrected denominator 3(N − 1) is unbiased under these assumptions.

$N \leq 1$ gives no finite-sample estimate of thermal variance after estimating the mean. Empty cells have no mean either.

Exercise: increase all velocities by (10, 0, 0) m/s. Answer: Ubar increases by that amount; both temperature estimates stay unchanged.

Averaging biased snapshot temperatures does not remove the bias. Correlated, transient, or unequal-weight samples need separate analysis.

Estimator reasoning before the cavity lab

The historical CPU mini-cavity remains a labeled bias-control exercise. The fixed-iid calculation is a conditional result, not a measurement of that cavity's bias.

A stationary sampling window can pool molecular moments and use a Bessel correction. This improves the estimator without proving independent samples.

Temporal correlation changes uncertainty and effective sample count. The correction does not remove startup transients, spatial discretization error, collision-model error, or wall-model error.

Self-check: distinguish estimator bias, statistical spread, and numerical/model bias before interpreting a temperature contour.

For a resolved comparison, retain particle count, time window, seed, time step, cell size, and estimator definition with every field.

See the Week 3 notebook for the retained CPU exercise and Project 5 for pooled-moment sampling.